

Summary

The plan addresses your goals:

✓ **Fixes duplicate question bugs** - Templates are selected deterministically based on conditions, can't duplicate by design

✓ **Improves question quality** - You control the template questions, AI only personalizes them

✓ **Maintains conversational personalization** - AI rewrites templates to reference character names naturally (e.g., "What drives Peter?" instead of "What drives your protagonist?")

Key Benefits:

- 70% cost reduction (gpt-4o-mini for simple personalization vs gpt-4o for full generation)
- 2-3x faster question generation
- Zero risk migration with feature flags
- No changes to UI or user-facing flows

The implementation creates 3 new files and modifies only one existing method in `openaiService.ts`, keeping all other features (answer analysis, "Surprise Me", Story Bible) intact.

Claude's Plan

Template-Based Questioning System for StoryBeats

Current State Analysis

Existing 4-Tier System

StoryBeats already has a tier-based questioning system built into the AI prompt in [openaiService.ts:338-363](#):

TIER 1 - CORE PROTAGONIST DETAILS:

- Protagonist identity/name
- Occupation
- Goal/desire
- Fears
- Motivations

TIER 2 - CONFLICT & ANTAGONIST:

- Main conflict/problem
- Antagonist/opposing force

TIER 3 - WORLD DETAILS:

- What makes world unique
- World rules and limitations

TIER 4 - SECONDARY ELEMENTS:

- Secondary characters
- Relationships
- Stakes

How Questions Work Now

1. **Dynamic AI Generation:** AI generates questions one at a time based on what's missing from Story Bible
2. **Adaptive Context:** Questions reference previous answers (e.g., "What drives [protagonist name]?")
3. **Smart Deduplication:** Prevents asking semantically similar questions
4. **Cost:** Each question requires an AI call (~1-2 cents per question)

Current Challenges

Based on recent commits:

- Fix for "stale data in interview question generation" (commit 56be5af)
- Fix for "asking for data already in Story Bible" (commit cdc0753)
- Fix for "repetitive questions" (commit 094cd14)

This suggests the AI-based system has reliability issues despite sophisticated duplicate prevention.

User's Proposal

Use predefined template questions for each tier instead of AI-generated questions.

Analysis: Template-Based vs AI-Generated

Benefits of Templates

1. **Consistency:** Same high-quality questions every time
2. **Reliability:** No risk of asking duplicate/stale questions
3. **Cost:** No AI calls needed for question generation
4. **Speed:** Instant question selection
5. **Control:** Easy to A/B test and refine questions

6. **Debugging:** Easier to trace issues

Tradeoffs of Templates

1. **Less Adaptive:** Can't reference user's specific story elements (character names, settings)
2. **Generic:** Questions might feel less conversational
3. **Rigid Ordering:** Harder to handle branching logic
4. **Edge Cases:** Need fallback logic for unusual stories

Potential Approaches

Option 1: Pure Template System

- Hardcode ideal questions for each tier
- Select next question based on Story Bible gaps
- Use string interpolation for personalization (e.g., "What drives {protagonist.name}?")

Pros: Simple, reliable, fast, cheap

Cons: Less flexible than current system

Option 2: Hybrid Template + AI Refinement

- Use templates as base questions
- Let AI personalize based on context
- AI only modifies wording, doesn't generate from scratch

Pros: Best of both worlds - consistent yet adaptive

Cons: Still requires AI calls (but simpler/cheaper)

Option 3: Template with Smart Selection

- Multiple template variations per tier
- Algorithm selects best template based on Story Bible state
- No AI needed for question selection

Pros: Fast, cheap, more variety than Option 1

Cons: Requires building sophisticated selection logic

Open Questions for User

1. **What problem are you trying to solve?**
 - Cost reduction?
 - Question quality/consistency?
 - Fixing the duplicate question bugs?

- All of the above?
- 2. **How important is personalization?**
 - Is it okay if questions don't reference character names/settings?
 - Or should we maintain that adaptive quality?
- 3. **Implementation preference?**
 - Pure templates (faster to build, less flexible)
 - Hybrid approach (more complex, better UX)
 - Template library with smart selection (middle ground)

Implementation Plan

Architecture: Template → Select → Personalize → Present

The hybrid system works in three stages:

1. **Template Selection** (deterministic) - Pick the right question based on Story Bible gaps and tier priority
2. **AI Personalization** (focused) - Make the template conversational using character names and story context
3. **Question Presentation** - Return fully structured Question object

Key Components

1. Question Templates (`/src/constants/questionTemplates.ts`)

New file containing 15-20 structured templates for all 4 tiers:

```
export interface QuestionTemplate {
  id: string;
  tier: 1 | 2 | 3 | 4;
  targeting: string; // Story Bible path
  templateText: string; // With {protagonist.name}
  placeholders
  requiredContext: string[]; // Fields needed for
personalization
  presentationMode: QuestionPresentationMode;
  chipOptions?: ChipOption[];
  allowSurpriseMe: boolean;
  condition?: string; // When to use this template
}
```

Example Templates:

- Tier 1: "What does {protagonist.name} do for a living?" (targets protagonist.occupation)

- Tier 2: "What is the main problem {protagonist.name} faces?" (targets conflict.mainConflict)
- Tier 3: "What makes the world of this story unique?" (targets world.description)
- Tier 4: "What happens if {protagonist.name} fails?" (targets conflict.stakes)

2. Template Selector (/src/services/ai/templateSelector.ts)

New service that deterministically selects the best template:

```
export class TemplateSelector {
  selectTemplate(bible: Partial<StoryBible>, history:
  QuestionHistory[]): QuestionTemplate | null {
    // Filter templates by:
    // 1. Condition evaluates to true (e.g., "protagonist.name
    exists AND occupation is empty")
    // 2. Required context available (or not needed)
    // 3. Not already asked

    // Sort by tier (ask Tier 1 before Tier 2)
    // Return first eligible template
  }
}
```

Condition Examples:

- "protagonist.name is empty AND protagonist.description is empty" → Ask identity question
- "protagonist.name exists AND protagonist.occupation is empty" → Ask occupation
- "conflict.mainConflict is empty" → Ask about main conflict

3. Question Personalizer (/src/services/ai/questionPersonalizer.ts)

New service that uses AI to make templates conversational:

```
export class QuestionPersonalizer {
  async personalizeQuestion(template: QuestionTemplate, bible:
  Partial<StoryBible>): Promise<Question> {
    // 1. Fill placeholders: {protagonist.name} → "Peter"
    // 2. Send to AI with simple prompt: "Make this
    conversational"
    // 3. AI rewrites: "What does Peter do for a living?"
    // → "So we've got Peter. What's his occupation?"
    // 4. Return Question object
  }
}
```

AI Prompt (simplified):

```
Template: "What does Peter do for a living?"  
Story Context: {protagonist: {name: "Peter", description: "nervous  
guy"}}
```

Make this question conversational and engaging while keeping the same intent.

Benefits:

- Uses gpt-4o-mini (10x cheaper than gpt-4o)
- Only 200-300 tokens per question (vs 500-800 for full generation)
- Simple task = better results
- Fallback to template text if AI fails

4. Integration with OpenAI Service

Modify: [openaiService.ts:265-295](#)

```
async generateSmartQuestion(  
  currentBible: Partial<StoryBible>,  
  questionHistory: QuestionHistory[]  
) : Promise<Question | null> {  
  // Stop after 10 questions  
  if (questionHistory.length >= 10) return null;  
  
  // NEW: Select template deterministically  
  const template = templateSelector.selectTemplate(currentBible,  
    questionHistory);  
  if (!template) return null;  
  
  // NEW: Personalize with AI  
  const question = await  
    questionPersonalizer.personalizeQuestion(template, currentBible);  
  
  // Keep duplicate check as safety net (should rarely trigger)  
  if (!this.isQuestionDuplicate(question.text, questionHistory,  
    currentBible)) {  
    return question;  
  }  
  
  return null;  
}
```

Keep unchanged:

- `analyzeResponse()` - Still uses AI to extract Story Bible data from answers
- `isQuestionDuplicate()` - Safety net (should rarely trigger now)
- "Surprise Me" feature - Still works
- `QuickInterviewScreen.tsx` - No changes needed

Migration Strategy

Phase 1: Parallel System (Low Risk)

1. Add new files (questionTemplates.ts, templateSelector.ts, questionPersonalizer.ts)
2. Add feature flag in openaiService.ts:

```
private useHybridQuestions = true; // Can toggle if issues arise
```
- 3.
4. Keep old `_generateSingleQuestion()` as fallback
5. Test hybrid system without breaking current functionality

Phase 2: Gradual Rollout

1. Deploy with feature flag enabled
2. Monitor for 1 week:
 - Duplicate question rate (should drop from ~15% to <2%)
 - Question quality (user skip rate)
 - Token usage (should drop ~70%)
 - Time to generate questions (should improve 2-3x)
3. If metrics good → keep
4. If issues → toggle flag back, fix, redeploy

Phase 3: Cleanup

1. Remove old `_generateSingleQuestion()` after 2 weeks
2. Clean up unused prompt engineering code
3. Update documentation

Why This Solves Your Problems

1. Fixes Duplicate Question Bugs ✓

Current Problem: AI generates semantically similar questions despite prompt rules

- "Who is your protagonist?" → "Can you describe your protagonist?"
- Duplicate detection is reactive, happens after wasteful generation

Hybrid Solution:

- Templates can't duplicate (each has unique ID and condition)
- Template selector checks `isAlreadyAsked()` proactively
- AI only personalizes, doesn't create question intent
- Duplicate detection becomes unnecessary safety net

2. Improves Question Quality ✓

Current Problem: 378-line prompt tries to enforce rules, AI may ignore

- Questions may be off-topic
- Tier sequencing is suggestion, not guarantee

Hybrid Solution:

- Every question has explicit targeting (Story Bible field)
- Tier system enforced programmatically (sort by tier)
- Guaranteed progression through character → conflict → world → secondary
- Templates vetted by you for quality

3. Maintains Personalization ✓

Current Problem: Fully AI-generated questions are generic

- May not reference character names naturally

Hybrid Solution:

- Templates have placeholders: "What does {protagonist.name} do for a living?"
- AI's sole job is making it conversational
- Focused prompt = better results: "What does Peter do for a living?" → "So we've got Peter as our protagonist. What's his occupation?"
- Character names always referenced when known

4. Cost & Performance ✓

Current System:

- Full generation: ~500-800 tokens per question
- Regeneration on duplicates: 2-3x cost
- Uses gpt-4o

Hybrid System:

- Personalization: ~200-300 tokens per question
- No regeneration needed
- Can use gpt-4o-mini (10x cheaper)
- Template selection: <1ms (instant)
- **Estimated 70% cost reduction**
- **2-3x faster question generation**

Critical Files to Modify

1. **CREATE** `/src/constants/questionTemplates.ts` - 15-20 templates covering all 4 tiers
2. **CREATE** `/src/services/ai/templateSelector.ts` - Template selection logic
3. **CREATE** `/src/services/ai/questionPersonalizer.ts` - AI personalization logic
4. **MODIFY** `/src/services/ai/openaiService.ts` - Update `generateSmartQuestion()` method
5. **NO CHANGES** `/src/screens/QuickInterviewScreen.tsx` - Consumes same Question interface

Testing Plan

Unit Tests:

- Template selector chooses correct template based on Story Bible state
- Template selector respects tier priority
- Condition evaluator works correctly
- Placeholder replacement works

Integration Tests:

- Complete interview flow from empty Bible to 10 questions
- Verify no duplicates occur
- Verify personalization includes character names
- Verify tier progression (1 → 2 → 3 → 4)

Manual Testing:

1. Fresh story (empty Bible) - verify asks for protagonist identity first
2. Partial story (name exists) - verify skips identity, asks occupation
3. Rich story (most fields filled) - verify only asks for gaps
4. Edge cases (missing name, special characters, long names)

Verification

After implementation, test end-to-end:

1. **Start Fresh Interview:**
 - Enter initial story idea in `InitialInputScreen`
 - Confirm format/genre/tone in `FormatConfirmationScreen`
 - Begin interview in `QuickInterviewScreen`
2. **Verify Question Flow:**
 - Question 1 should ask for protagonist identity (Tier 1)
 - Answer with character name

- Question 2 should reference that name and ask for occupation (Tier 1)
- Continue through ~7 questions
- Verify progression: Protagonist → Conflict → World → Secondary
- 3. **Check Duplicate Prevention:**
 - Review all questions asked
 - Confirm no semantic duplicates
 - Confirm no questions about data already in Story Bible
- 4. **Check Personalization:**
 - After providing protagonist name, all questions should reference it
 - Questions should feel conversational, not robotic
- 5. **Check Story Bible:**
 - After interview, inspect Story Bible JSON
 - Verify all fields populated correctly
 - Verify no data loss or overwriting
- 6. **Test "Surprise Me":**
 - Click "Surprise Me" on a question
 - Verify generates 4 AI answer options
 - Verify Story Bible updates correctly after selecting option

Success Criteria

Deploy is successful if:

- ✓ Zero duplicate questions in 10 test interviews
- ✓ All questions reference protagonist name when known
- ✓ Questions progress through tiers 1 → 2 → 3 → 4
- ✓ Story Bible fully populated after interview
- ✓ Token usage reduced by 60-70%
- ✓ Question generation time <1 second
- ✓ No regressions in answer analysis or "Surprise Me" features

Stayed in plan mode

- 1.
- 2.
- 3.

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Excellent! I understand - you want to provide the foundational structure first before we finalize the plan. This makes perfect sense.

So the Story Bible is not just for storing answers to specific questions, but rather:

- **Opportunistic context capture** - Extract ALL relevant info from user's initial idea and ongoing answers
- **Pattern recognition** - AI should identify connections and story elements the user might not see
- **Creative constraint engine** - Give AI rules to work within, then let it surprise you with connections

And you're going to provide:

1. **The 12 beats definition** - What each beat requires
2. **Tiered questions mapped to beats** - Which questions gather info for which beats
3. **Information requirements per beat** - What Story Bible data is needed to complete each beat

This will help us design the template questions to:

- Target specific beats (not just generic Story Bible fields)
- Extract the right information to enable beat completion
- Give AI enough context to spot interesting patterns

I'm ready for the detailed beat definitions and question mappings. This will significantly improve the plan by tying the questioning system directly to your 12-beat story structure!